© 2016 Human Kinetics, Inc.

Note. This article will be published in a forthcoming issue of the *International Journal of Sports Physiology and Performance*. The article appears here in its accepted, peer-reviewed form, as it was provided by the submitting author. It has not been copyedited, proofread, or formatted by the publisher.

Section: Original Investigation

Article Title: A New Method to Individualize Monitoring of Muscle Recovery in Athletes

Authors: Anne Hecksteden[1], Werner Pitsch[2], Ross Julian[1], Mark Pfeiffer[3], Michael Kellmann[4,5], Alexander Ferrauti[4] and Tim Meyer[1]

Affiliations: [1]Institute of Sports and Preventive Medicine, Saarland University, Saarbruecken, Germany. [2]Institute for Sport Sciences, Dept. Sociology and Economy of Sports, Saarland University, Saarbruecken, Germany. [3]Institute of Sports Science, Johannes-Gutenberg University, Mainz, Germany. [4]Faculty of Sports Science, Ruhr-University of Bochum, Bochum, Germany. [5]Schools of Human Movement Studies and Psychology, The University of Queensland, Queensland, Australia.

Journal: *International Journal of Sports Physiology and Performance*

Acceptance Date: November 28, 2016

©2016 Human Kinetics, Inc.

DOI: http://dx.doi.org/10.1123/ijspp.2016-0120

"A New Method to Individualize Monitoring of Muscle Recovery in Athletes" by Hecksteden A et al.
International Journal of Sports Physiology and Performance
© 2016 Human Kinetics, Inc.

Original Investigation

A new method to individualize monitoring of muscle recovery in athletes

Anne Hecksteden[1], Werner Pitsch[2], Ross Julian[1], Mark Pfeiffer[3], Michael Kellmann[4,5], Alexander Ferrauti[4] and Tim Meyer[1]

[1] Institute of Sports and Preventive Medicine, Saarland University, Saarbruecken, Germany

[2] Institute for Sport Sciences, Dept. Sociology and Economy of Sports, Saarland University, Saarbruecken, Germany

[3] Institute of Sports Science, Johannes-Gutenberg University, Mainz, Germany

[4] Faculty of Sports Science, Ruhr-University of Bochum, Bochum, Germany

[5] Schools of Human Movement Studies and Psychology, The University of Queensland, Queensland, Australia

Corresponding author: Dr. Anne Hecksteden

Institute of Sports and Preventive Medicine,

Saarland University,

Campus Building B8.2

66123 Saarbrucken, Germany

Email: a.hecksteden@mx.uni-saarland.de

Phone: 0681-302 70400 Fax: 0681-302 70400

Running head: Individual ranges for markers of muscle recovery

Abstract word count: 250
Text only word count: 3906
Figures: 1
Tables: 4

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

ABSTRACT

Purpose: Assessment of muscle recovery is essential for the daily fine-tuning of training load in competitive sports, but individual differences may limit the diagnostic accuracy of group-based reference ranges. We report an attempt to develop individualized reference ranges using a Bayesian approach comparable to that developed for the athlete biological passport. **Methods:** Urea and creatine kinase (CK) were selected as indicators of muscle recovery. For each parameter, prior distributions and repeated measures standard deviations were characterized based on data of 883 squad athletes (1758 data points, 1-8 per athlete, years 2013-2015). Equations for the individualization procedure were adapted from previous material to allow for discrimination of 2 physiological states (recovered; non-recovered). Evaluation of classificatory performance was carried out using data from 5 consecutive weekly microcycles in 14 elite junior swimmers and triathletes. Blood samples were collected every Monday (recovered) and Friday according to the repetitive weekly training schedule over five weeks. On the group level, changes in muscle recovery could be confirmed by significant differences in urea, CK and validated questionnaires. Group-based reference ranges were derived from that same dataset to avoid overestimating the potential benefit of individualization. **Results:** For CK error rates were significantly lower with individualized classification (p vs. group-based: test-pass error rate: p=0,008; test-fail error rate: p<0,001). For urea numerical improvements in error rates failed to reach significance. **Conclusions:** Individualized reference ranges seem to be a promising tool to improve accuracy of monitoring muscle recovery. Investigating application to a larger panel of indicators is warranted.

Keywords: reference range, distribution, individualization, sport, recovery

© 2016 Human Kinetics, Inc.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

INTRODUCTION

The assessment of fatigue and recovery can be essential for the daily fine-tuning of training load in competitive sports. Blood born markers are objective, have low technical error of measurement, the necessary sampling interferes only minimally with the training process and their physiological relation to the different aspects of exercise-induced physical fatigue is well understood. However, as for other fatigue indicators, the variability of measured values as well as of their fatigue-induced changes is large[1-3]. This leads to wide reference ranges, even when based on athletic populations, and thereby severely impedes on the diagnostic value of all parameters known to date[3]. In search for a solution to this dilemma it is important to note that reference ranges are based on cross-sectional data and therefore include intra- as well as interindividual variation. Elimination of the interindividual component by deriving reference ranges from repeated measurements in a given athlete may plausibly lead to smaller, individualized reference ranges and consecutive improvements in diagnostic accuracy.

The practical relevance of this consideration is supported by the substantial proportion of interindividual variation in common fatigue markers[2]. However, a considerable number of individual measurements will be needed to realize this theoretical advantage due to the connection between degrees-of-freedom and standard error. Therefore, the gradual individualization of a population-based prior distribution by successive inclusion of information from individual measurements seems to be the most promising solution. This "Bayesian" strategy, which is widely used in forensic science[4], is also the idea underlying the athlete biological passport[5,6] as implemented by the World Anti-Doping Agency (WADA). However, in the case of fatigue assessment, two physiological states have to be discriminated and interindividual variation seems to be present with regard to the general "level" of indicators (between-subject variation) as well as concerning the magnitude of fatigue induced

© 2016 Human Kinetics, Inc.

changes (fatigue-by-person interaction). Therefore, decisions should likely be based on two reference ranges (fatigued and recovered) instead of simply detecting deviations from the recovered state.

The present study aims to explore the adaptation and application of the approach explained above for the assessment of muscle recovery in competitive athletes using creatine kinase (CK) and urea (U) as biomarkers. To fully utilize its potential, a three-step approach was implemented:

(1) Characterization of sport-specific prior distributions and reference ranges based on routine measurements from 883 squad-athletes.

(2) Development of the individualization algorithm as an extension to previous work allowing for decisions based on reference ranges for two recovery states.

(3) Evaluation using measurements from five consecutive weekly microcycles in high-level athletes.

To improve transparency and facilitate further development and application, care has been taken to reduce the complexity of mathematical notation as compared to previous work[5]. Moreover, an Excel spreadsheet is available (supplemental digital content 1) to enable convenient testing. Documented statistical code is provided in supplemental digital content 2.

METHODS

General remark

The purpose of the following method development is to create individualized cut-off values to distinguish between two states of muscle recovery ("recovered" (R) and "non-recovered" (NR)). For this purpose, knowledge about a relevant reference population (in the following, the priori distribution) will be combined with repeated individual measures from R and NR time points, respectively, to generate individualized posteriori distributions. This proceeding exemplifies the rationale of Bayesian statistics. Cut-off values are then calculated based on the individualized posterior distributions for R and NR states, respectively. Figure 1

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

displays the development of 95% confidence intervals over the gradual individualization procedure together with the measured values for one male subject (male athlete 1 in supplemental material 4). The complete set of courses for the chronological order of values is provided in supplementary material 4 (males) and 5 (females).

Characterization of sport-specific prior distributions

Despite the frequent and long-standing use of CK and urea as markers of muscle recovery, specific information about their distribution in athletic populations is scarce[7-9]. The same gap in knowledge applies for intraindividual (repeated measures; RM) variation, estimates of which are needed to compute individualized posterior distributions. Therefore, prior distributions, to be used as starting points for the individualization procedure, as well as RM-standard deviations were considered an integral part of this project.

The respective analyses were based on anonymized data from all routine examinations of squad-athletes between January 2013 and November 2015 conducted at the Institute of Sports and Preventive Medicine. The following sports were represented with more than 10 individuals (in alphabetical order): cycling, team sports, rowing, swimming, track-and-field, and triathlon. Data of 52 athletes from various other sports were pooled with the team sports for analysis of eventual sport-specific differences. In total, 1758 data points (♀ 509; ♂ 1249) from 883 (♀ 210; ♂ 673) individuals could be included for either parameter.

In order to account for the intraindividual variance within the dataset and to best exploit the available information, a leave-1-out Jackknife resampling[10] was conducted for subjects, irrespective if there were only one or multiple records for each omitted subject. For subjects with multiple records in the Jachknife sample, only one record was randomly drawn thus excluding intraindividual variance from the resampled datasets. In the following, we will call this resampling method lmo-Jackknife (for "leave multiple records out").

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

Preliminary analyses demonstrated significant differences in means between males and females as well as high skewness of distributions for both sexes (cp. supplemental digital content 3). Therefore, further analyses were conducted separately for males and females using log-transformed data. In the logic of a classical Jackknife estimation, the relevant distribution parameters as well as their confidence intervals were estimated by calculating the mean of these parameters in the lmo-resamples.

For 132 males and 87 females with repeated measurements (RM), data were additionally analysed in order to obtain an estimator of RM-variation. The weighted individual standard deviations were calculated as the square root of the summed variances, weighted by the degrees of freedom of the number of individual measurements.

Development of the individualization algorithm

Prior distributions $P(m_0, s_0^2)$ and repeated measures standard deviations (s_{RM}) for males and females, derived as described above, were used as starting point for the individualization procedure.

Individualization by successive inclusion of information from values measured in the individual athlete (x) was conducted using equation 1. Thereby the mean resulting from the first individualization step (m_1) is the average of the priori mean (m_0) and the observation (x_1) weighted for the relationship between the interindividual variance in the prior distribution (s_0^2) and mean repeated measures variance (s_{RM}^2). For each following step, the posteriori mean (m_{n+1}) is derived from the mean from the preceding distribution mean (m_n) and the observation (x_{n+1}) Moreover, respecting the Bayesian rationale of this approach, the posterior standard deviation of the preceding step (s_n; cp equation 2 below) is then used instead of s_0.

(1) $$m_{n+1} = \frac{s_{RM}^2 * m_n + s_n^2 * x_{n+1}}{s_{RM}^2 + s_n^2}$$ equation 5.7 in [4]

© 2016 Human Kinetics, Inc.

Individualization of s_0 was based on equation 2 (equation 5.10 in[4]). According to a strict Bayesian logic, s_n will necessarily decrease with increasing number of individual measurements. Although we assume this characteristic implausible from a physiological perspective, a comparison between using individualized repeated measures standard deviations or standard deviations which were obtained ex ante from the priori distribution analysis, respectively revealed no advantage of individualized repeated measures standard deviations with respect to classificatory performance. Therefore, respecting the law of parsimony, the strictly Bayesian approach was implemented.

$$s_{n+1} = \sqrt{\frac{1}{(n+1) \times \frac{1}{s_{RM}^2} + \frac{1}{s_0^2}}} \quad \text{(equation 5.10 in[4])} \tag{2}$$

Upper and lower limits of individual reference ranges (figure 1; supplemental digital content 4 and 5) were generated by transforming results of the individualization back to the original scale. All steps were conducted separately for male and female subjects due to the different characteristics of the respective prior distributions (tables 1 and 2). Classification results for males and females were then pooled for assessment of error rates.

Classification

Based on the results of the final individualization step a cut-off value was calculated from the posteriori distributions for values from NR ($m_{4\text{-}NR}$) and R ($m_{4\text{-}R}$) time points, respectively. Cut-off was set at equal distances from the two posterior means ($m_{n\text{-}NR}$ and $m_{n\text{-}R}$).

For urea as well as for CK high values indicate a lesser state of muscle recovery. However, despite the significant difference between means, in some cases $m_{4\text{-}NR}$ may be smaller than $m_{4\text{-}R}$. In these cases a meaningful classification of muscle recovery status would

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

not be possible based on that parameter. Consequently, no individualized classification was conducted.

Empirical data

Test-runs of the individualization procedure were based on data from five successive, weekly microcycles in 14 junior elite athletes (triathletes and swimmers, nine male, four female) recruited at the federal Olympic Training Centre Saarbrücken. According to the repetitive weekly microcycles during the general preparation period, venous blood samples were collected each Monday (recovered after day off on Sunday (R)) and Friday (after four days of strenuous training (NR)) morning. Prior to blood collection, each participant completed the a validated questionnaire (Acute Recovery and Stress Scale[11]; ARSS) to verify effective induction of perceived changes in recovery status. Adherence to the training schedule was checked for every individual by reviewing training logs filed by the responsible coach. Repetitive performance tests were not acceptable for the enrolled high-level athletes and their coaches. Only cases without missing data were included in this analysis. An outline of the respective training programs as well as sampling and analytical methods have been described previously[2]. The study was approved by the institutional review board (Aerztekammer des Saarlandes, approval number 228/13).

Individualization of distributions for R and NR time points, respectively were conducted using log-transformed data. According to the number of microcyles, the dataset comprised five R and five NR time points per person. In some cases, Monday readings for CK were higher than for the preceding Friday, presumably due to uncontrolled leisure-time physical activities. In this case, the Monday value was excluded from analysis thus ensuring that only valid data were used. In total 8 values were excluded. Consequently, four non recovered and up to four recovered values were used for individualization and the remaining value was classified according to the posterior distribution resulting from the last

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

individualization step (m_4 and s_4). To enable classification of all values, five individualization runs were conducted per person, each starting with the sample included second in the run before (run 1: individualization using samples 1 to 4 in chronological order followed by classification of sample 5; run 2: individualization using samples 2 to 5 followed by classification of sample 1, run 3: individualization using samples 3 to 5 and 1 followed by classification of sample 2 and so forth).

Evaluation of classificatory performance

Performance of the individualization procedure was assessed with respect to error rates in classifications of data points as R and NR, respectively. Test-pass error rate was defined as the proportion of data points, falsely classified as R among all R classifications. Test-fail error rate was defined as the proportion of data points falsely classified as NR among all NR classifications.

To enable a comparative interpretation of error rates, we also classified the data points using an alternative group-based approach, based on the mean and standard deviation of NR and R time points in this same dataset. Likewise, to the individualized approach, cut-off values were set at equal distances from the group means (m_{fat} and m_{rec}). Male and female subsamples were analysed separately.

Performance of the classification was compared between the individualized and the group based classification method in an exact binomial test. We interpreted the error rates as error probabilities and tested the hypothesis that for CK (urea) the probability of false classifications with individualized cut-off-values was lower than when using group-based cut-off values. The tests were conducted for NR (test-fail errors) and R data points (test-pass errors) as well as for CK and urea independently.

Due to the low number of observations in our sample, we estimated the CIs for the test-pass and test-fail error rates (see table 4) from Monte-Carlo simulations. We will report

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

the 2.5th and 97.5th percentile of the distribution of the simulation results. The Monte Carlo runs started with simulated samples with normally distributed sample means and normally distributed repeated measures. Means, inter- and intra-individual standard deviations from the male subsample in our empirical data were used to create the Monte-Carlo samples. The female subsample was assumed to be too small to provide reliable estimators. For 1000 samples, each starting with 100 individuals, we conducted the same calculations of individualized posteriori distributions and subsequent and classifications as described for our empirical sample. Details of the Monte Carlo simulation as well as the used R-Code are provided in supplement digital content 2.

RESULTS

Group-based prior distributions and repeated measures variation

Distribution type

Relevant distribution parameters are provided in table 1. High skewness and kurtosis support the assumption of lognormality in all four cases (CK and urea in males and females, respectively).

Comparison between different sports

The analysis by disciplines showed an unequal distribution of both genders by sports discipline. For male subjects, the (lmo-jackknife) t-statistics showed no significant difference for CK (t=-0.025, p=0.98) and urea (t=0.811, p=0.417). For female subjects, the results were similar (for CK: t=0.212, p=0.862, for urea: t=1.396, p=0.164). By and large, these results indicated that there was no need for a sport-specific estimation of the prior distributions.

Repeated measures variation

Weighted mean variances and standard deviations are shown in table 2.

Performance of the individualization procedure

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Mean values for urea and CK were significantly higher for the NR as compared to the R time points (urea: R 31,6 ± 4,8 mg/dl; NR 35,3 ± 6,9 mg/dl; Δ 3,7 ± 4,3 mg/dl p=0,001; CK: R 198 ± 92 U/l; NR 259 ± 118 U/l; Δ 61 ± 96 U/l p=0,033). Likewise, all 8 dimensions of the ARSS showed according significant differences.

Test-pass and test-fail error rates for the group of 14 athletes are summarized in table 3. Numerically, for both parameters test-pass and test-fail error rates were lower with individualized categorization as compared to group-based cut-off values. Improvements in error rates were statistically significant for CK (test pass: $p<0.001$; test-fail $p=0.008$). For urea the respective differences failed to reach statistical significance.

Confidence intervals for the error rates in male subjects are detailed in table 4. For female subjects no simulation was conducted due to the low number of subjects. According to the method, confidence intervals were not perfectly symmetrical relative to the median. The confidence intervals for the four error rates had ranges from 0.162 (test-pass error rate for urea) up to 0.216 (test-fail error rate for CK). The empirical test-fail error rate for CK (0.074, both sexes) did not fall within the confidence interval as derived from Monte Carlo simulations (lower limit 0.087, males only). For the other error rates, the confidence intervals covered the empirical values.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

DISCUSSION

Reference ranges are typically based on data from a large number of subjects and therefore reflect the distribution of the respective parameter in a certain population. However, if the typical value of a biomarker varies widely between different persons, as it is the case with fatigue indicators[1,3], this group-based approach leads to reference ranges which are inappropriately wide for a given subject. This impedes on the detection of values which are remarkable for a specific person (albeit not with respect to the population). By contrast, fatigue assessment in high-level sports aims at the accurate detection of subtle differences in

© 2016 Human Kinetics, Inc.

the individual athlete. The Bayesian approach presented in this work addresses the above problem by enabling an individualized assessment of muscle recovery. The rationale is similar to the athlete biological passport[5] as implemented by the World Anti-Doping Agency (WADA): The individualization procedure starts with a sport-specific prior distribution and successively integrates information from individual measurements to progress towards individualized distributions for R and NR values. Based on these posterior distributions, individualized reference ranges and cut-off values may be calculated. In a repeated measures dataset from 14 elite junior athletes individualized reference values for CK performed significantly better compared to a group-based classification. For urea differences in the same direction failed to reach statistical significance. Confidence intervals for test-pass and test-fail error rates derived from Monte Carlo simulations were reasonably narrow. This result is also in support of actual differences in diagnostic accuracy between individualized and group-based classification.

The aim of this trial was to investigate the potential advantage of individualized as compared to group-based reference values in the assessment of muscle recovery. In order to enable such an inference, the choice of the comparative group-based reference is crucial. In particular, confounding the effect of individualization with possible advantages of prior distributions derived from closely matched groups of athletes could lead to overestimating the effect of individualization. To avoid this, the group-based reference ranges used in this trial were based on means and standard deviations of the same data to be subsequently classified. By contrast, prior distributions for the individualization procedure were derived from an independent sample of athletes from different sports. The individualization process was constructed conservatively involving a rather low number of individual observations. Obvious outliers, specifically CK- and urea-values from recovered states which exceeded the value of the preceding non-recovered state, were excluded from further analysis

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

(individualized and group-based). Due to the measurement sequence which started in the recovered state, this did not hold for the first recovered time point. In future studies as well as with regards to potential practical application, the measurement regime should be changed and start at a non-recovered state. This may increase the advantage of individualization considerably[2].

Error rates resulting from the algorithmic, dichotomous classification of measured values provide objective support for the advantage of individualized assessment. However, graphical representations of measured values together with the gradual development of reference ranges over successive individualization steps offer a wealth of additional information (cp. figure 1 as well as supplemental digital content 4 and 5). First and most importantly, these graphs can be used as an intuitively comprehensible plausibility check for the individualization procedure and any classification. This is an important advantage over purely abstract, higher-order classifiers, in particular with regard to the potential transfer into sports practice. Moreover, these graphs illustrate that the "success" of individualization for a particular indicator differs widely between individuals depending on the difference between individual and group-based distributions. An individualization of reference ranges is of course particularly beneficial in cases with habitual readings which deviate clearly from the group-based means. Finally, the gradual separation of reference ranges and the performance of cut-off values will depend on the mean difference between R and NR values as well as on the variability of measurements within both physiological states. This "individual effect size" finally determines the diagnostic potential for a given indicator and athlete. In cases where the individual effect size is low, individualized assessment cannot recover discriminatory power which is inexistent in the first place. This becomes particularly apparent in cases where the relationship between R and NR values is repeatedly in clear contrast to the main

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

effect (e.g. supplemental digital content 4 h and r as well as supplemental digital content 5 b and h).

The core of the approach presented in this work consists in the stepwise individualization of distributions for R and NR values of the respective parameter. These posterior distributions are subsequently used to derive individualized reference ranges or cut-off values. While such indicators are needed for practical decision making, it is important to keep in mind that there is no compulsory way to derive them. Obviously, meaningful cut-off values will be located between posterior means for recovered and non-recovered states. In this work, the cut-off value is placed at equal distances from the two posterior means. However, alternative cut-off values may be appropriate (and admissible) depending on the framework conditions e.g. the cut-off may be shifted towards the posterior mean for R values in order to favor sensitivity in the detection of fatigue. Moreover, while dichotomous classifications are needed to quantify sensitivity and specificity in scientific work, other ways of rating may reflect the gradual character of fatigue and recovery more accurately (e.g. ratio of quantiles for the non-recovered and recovered distributions or likelihood of fatigue).

The individualization procedure has been developed using CK and urea as examples from a wide range of potential indicators. Beyond the generally favorable test quality criteria of blood born markers (low technical error), CK and urea were used because they are longstanding, well studied and practically relevant indicators of muscle recovery[3]. However, it is important to keep in mind that exercise induced fatigue is a complex phenomenon involving many domains which are not represented by these markers (e.g. hormonal, autonomic, central nervous, immunological, cardiocirculatory). In principle, the method reported here may be transferred to other parameters within and beyond the field of monitoring recovery in competitive athletes. Considering the need for repeated measurements and aiming at transfer into sports practice, exploring non-invasive markers such as resting

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

heart rate, heart rate variability or even subjective measures seems particularly promising. However, care has to be taken on following the whole process starting with an estimation of inter- and intraindividual variance components and characterization of prior distributions in a large sample of athletes and ending with an evaluation of diagnostic performance. Moreover, generalization of the individualization procedure to multivariate distributions may enable an integrative rating of overall fatigue status. This continuative consideration is supported by the recent report of consistent individual patterns of exercise induced changes in recovery status in elite athletes from one training group[2]. Consequently, follow up studies should include a large and well-considered panel of indicators because characterization of multivariate prior distributions requires combined measurement of all parameters to be potentially included.

In this trial, the criterion classification of time points as "recovered" or "non-recovered" was conducted using convergent evidence on the group (training schedules, validated questionnaires, blood born fatigue markers) as well as on the individual level (training logs). While in principle a single criterion measure would be desirable, a valid and practicable instrument, reflecting muscle recovery objectively and accurately is not available. Therefore, despite the clear contrast in muscle recovery as evidenced by training load and group based analysis of psychometrical and blood-based markers, misclassification of individual time points may not be ruled out completely. However, such misclassifications on the criterion level, if present, would concern individualized and group based approaches equally and may be assumed to induce overestimation of error rates.

Confidence intervals for test-pass and test-fail error rates (table 4), respectively, were estimated by Monte Carlo simulations. Simulations were based on the distribution characteristics of male subjects. No simulation was conducted for female subjects due to the low number of female athletes. This has to be taken into account when comparing confidence limits to the empirically observed error rates which are based on data from male and female

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

athletes. Specifically, the empirical test-fail error rate for CK did not fall within the confidence interval which was derived from the Monte Carlo simulations. Further empirical results, especially from female athletes may give a hint if this effect was sample-specific for our study or if it resulted from gender difference between the empirical analysis and the simulations or from systematic effects in the empirical data (e.g. autocorrelation) which was not taken into account in our simulation. Nevertheless, the empirical analysis has resulted even better in terms of error rates as we would have expected from the result of the simulation.

PRACTICAL APPLICATION

Large interindividual variation severely impedes on the practical value of biomarkers for muscle recovery when rating the status of an individual athlete. The presented approach enables the individualization of reference ranges and thereby leads to improvements in diagnostic accuracy. Moreover, graphical representations offer intuitive plausibility control as well as insights into individual effect sizes and time trends. Therefore:

- Individualized reference ranges (and other indicators derived from individualized posterior distributions) can be used to optimize accuracy of assessment in competitive athletes.
- Visualization of raw data together with the development of reference ranges over repeated measurements may be used by athletes and coaches to control and comprehend the individualization and decision process. This is an important advantage over abstract, higher-order classifiers.
- "individual effect sizes" (difference between R and NR posterior means and intraindividual variation) may be used to individualize parameter panels. This may lead to clearer decisions, reduced burden for the athlete and lower employment of resources.

CONCLUSIONS

Individualizing reference ranges of biomarkers for muscle recovery offers a promising tool to improve accuracy of monitoring in competitive athletes. Practical application as well as transfer to other, ideally non-invasive, parameters warrants to be explored in future studies.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

ACKNOWLEDGEMENTS

The present study was funded by the German Institute of Sport Science. The research was realized within the project "RegMan - Management of Regeneration in Elite Sports" (IIA1-081901/12-16).

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

REFERENCES

1. Hecksteden A, Skorski S, Schwindling S, Hammes D, Pfeiffer M, Kellmann M, Ferrauti A, Meyer T. Blood-Borne Markers of Fatigue in Competitive Athletes - Results from Simulated Training Camps. *PLoS One.* 2016;11(2):e0148810.

2. Julian R, Meyer T, Fullagar HH, Skorski S, Pfeiffer M, Kellmann M, Ferrauti A, Hecksteden A. Individual patterns in blood-borne indicators of fatigue - trait or chance. *J Strength Cond Res.* Mar 1 2016.

3. Meeusen R, Duclos M, Foster C, Fry A, Gleeson M, Nieman D, Raglin J, Rietjens G, Steinacker J, Urhausen A. Prevention, diagnosis, and treatment of the overtraining syndrome: joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. *Med Sci Sports Exerc.* Jan 2013;45(1):186-205.

4. Aitken C.G.G., F. T. *Statistics and the Evaluation of Evidence for Forensic Scientists*. 2 ed: John Wiley & Sons; 2004.

5. Sottas PE, Baume N, Saudan C, Schweizer C, Kamber M, Saugy M. Bayesian detection of abnormal values in longitudinal biomarkers with an application to T/E ratio. *Biostatistics.* Apr 2007;8(2):285-296.

6. Sottas PE, Vernec A. Current implementation and future of the Athlete Biological Passport. *Bioanalysis.* Jul 2012;4(13):1645-1652.

7. Lev EI, Tur-Kaspa I, Ashkenazy I, Reiner A, Faraggi D, Shemer J, Argov Z. Distribution of serum creatine kinase activity in young healthy persons. *Clin Chim Acta.* Jan 1999;279(1-2):107-115.

8. Mougios V. Reference intervals for serum creatine kinase in athletes. *Br J Sports Med.* Oct 2007;41(10):674-678.

9. Nicholson GA, McLeod JG, Morgan G, Meerkin M, Cowan J, Bretag A, Graham D, Hill G, Robertson E, Sheffield L. Variable distributions of serum creatine kinase reference values. Relationship to exercise activity. *J Neurol Sci.* Dec 1985;71(2-3):233-245.

10. Efron B. Nonparametric estimates of standard error: The jackknife, the bootstrap and other methods. *Biometrica.* 1981;68(3):589-599.

11. Kölling S, Hitzschke B, Holst T, Ferrauti A, Meyer T, Pfeiffer M, Kellmann M. Validity of the Acute Recovery and Stress Scale – Training Monitoring of the German Junior National Field Hockey Team. *Int J Sports Sci Coach.* 2015;10(2-3):529-542.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

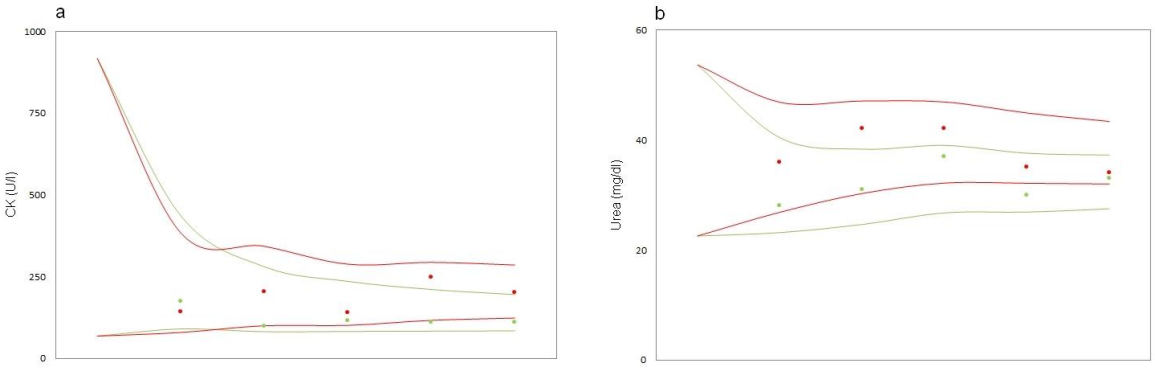

Figure 1. Development of individualized reference ranges over five weekly microcycles (a: creatine kinase; b: urea)

Boundaries represent 95% confidence intervals

Dots represent measured values

Green: recovered Red: non-recovered

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Table 1. Prior distribution – Descriptive log'ed and standard statistics

	male		female	
	logCK	**logUrea**	**logCK**	**logUrea**
Median	5,462	3,555	5,077	3,341
Mean	5,527	3,550	5,101	3,334
SD	0,661	0,221	0,597	0,254
skewness	0,598	-0,135	0,469	0,102
kurtosis	0,337	0,381	0,550	0,650
	CK (U/l)	**Urea (mg/dl)**	**CK (U/l)**	**Urea (mg/dl)**
Median	236	35	160	28
Mean	251	35	164	28
SD	2	1	2	1

Estimated via leave multiple records out (lmo-) Jackknife-method

Standard statistics were derived by reverse transformation of log'ed values.

CK: creatine kinase; SD: standard deviation

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Table 2. Repeated measures variation

	male		female	
	logCK	**logUrea**	**logCK**	**logUrea**
Mean	5.575	3.602	5.121	3.388
SD	0.504	0.185	0.489	0.186
	CK (U/l)	**Urea (mg/dl)**	**CK (U/l)**	**Urea (mg/dl)**
Mean	264	37	168	30
SD	2	1	2	1

Standard statistics were derived by reverse transformation of log'ed values.

CK: creatine kinase; SD: standard deviation

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Table 3. Test pass error rate and test fail error rates for individualized and group-based classification.

	Individualized	Group-based	p
Urea			
Test-pass error rate	0.286	0.304	0.637
Test-fail error rate	0.250	0.279	0.601
CK			
Test-pass error rate	0.180	0.333	0.008
Test-fail error rate	0.074	0.317	<0.001

Test-pass error rate: proportion of data points, falsely classified as recovered

Test-fail error rate: proportion of data points, falsely classified as non-recovered

CK: creatine kinase

p is reported for exact binomial tests, using the group based error rates as probabilities for the binomial distribution and calculating the probability to get the number of errors with individualized cut-off values in 62 (CK) resp. 44 (urea) decisions.

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Table 4. Confidence Interval and Median for error rates in Monte-Carlo simulations with individualized cut-off values.

	2.5th	MD	97.5th
Urea			
Test-pass error rate	0.218	0.303	0.389
Test-fail error rate	0.131	0.229	0.347
CK			
Test-pass error rate	0.145	0.224	0.307
Test-fail error rate	0.081	0.160	0.258

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0

© 2016 Human Kinetics, Inc.

Supplemental digital content

Supplemental digital content 1	Excel spreadsheet implementation of the individualization procedure
Supplemental digital content 2	Detailed description of simulations and statistical code
Supplemental digital content 3	Preliminary analyses for deduction of prior distributions
Supplemental digital content 4	Development of individualized reference ranges for male athletes Boundaries represent 95% confidence intervals Dots represent measured values Green: recovered Red: non-recovered
Supplemental digital content 5	Development of individualized reference ranges for female athletes Boundaries represent 95% confidence intervals Dots represent measured values Green: recovered Red: non-recovered

Downloaded by t_lambrinides@yahoo.com on 10/16/17, Volume 0, Article Number 0